

CHINTA P K SEKHAR REDDY

SOFTWARE ENGINEER (CLOUD & DEVOPS FOCUS) | CLOUD & INFRASTRUCTURE OWNER

Hyderabad, India

☎ +91 93981 70513

✉ pksekharreddychinta@gmail.com

🌐 p-k-sekhar-reddy-chinta

PROFESSIONAL SUMMARY

DevOps & Cloud Engineer with 2.5+ years of hands-on experience independently architecting, deploying, and operating production infrastructure across 4 concurrent platforms in the EV, travel, and fintech domains, as the sole engineer responsible for infrastructure, cloud, and DevOps end-to-end. Own the complete infrastructure stack solo — AWS & Vultr multi-cloud, on-premises servers, CI/CD pipelines, container orchestration, Infrastructure as Code (Terraform, Ansible), Apache Kafka event streaming, Redis, observability (Prometheus, Grafana, cAdvisor, Node Exporter), and security hardening (AWS Cognito, RSA-based API protection) — while also contributing hands-on to Java Spring Boot backend REST API design, running at production scale with real-money transactions.

- \$1,300+/mo in verified infrastructure cost savings — 83% AWS cost reduction (\$1,200 → \$200/mo via hybrid on-prem design) plus \$300/mo eliminated by migrating from Confluent to self-hosted Apache Kafka.
- API performance — Reduced key API response times from ~700ms to under 200ms via caching, DTO projection, and response compression.
- Protocol & real-time systems — Implemented OCPP 1.6J over WebSockets for real-time EV charging station communication.
- 99.9%+ uptime maintained across all production platforms via self-healing scripts, CloudWatch alerting, and Spring Boot Actuator health checks.

TECHNICAL SKILLS

Cloud – AWS: EC2, S3, RDS, ECS (Fargate & EC2), ECR, ALB, Route 53, CloudWatch, IAM, ACM, WAF, Cognito (OAuth 2.0/JWT, API protection), SNS, SES, Auto Scaling

Cloud – Vultr: Instances, Firewall Groups, Load Balancers, Block Storage, VPC Networking

Infrastructure as Code & Automation: Terraform, Ansible (AI-assisted implementation using Claude for rapid, fast-paced delivery), HashiCorp Vault (secrets management)

DevOps: Docker, Portainer, Linux/Ubuntu Server, Windows, PowerShell, Nginx, Shell Scripting, Cloudflare Tunnel (cloudflared), Zero-Trust Networking, SSL/TLS, SSH Key Auth

Backend: Java, Spring Boot, Spring Security, REST API Design, Microservices Architecture, JWT, OAuth 2.0, WebSockets, OCPP 1.6J, OCPI (in progress), Maven, JavaMail/SMTP, Razorpay Payments API, RSA Asymmetric Encryption, Node.js

Streaming & Caching: Apache Kafka (self-hosted), Topic Management, Real-Time Event Streaming, Confluent Kafka Migration, Redis (containerized microservice)

CI/CD: Jenkins (Master-Slave), GitHub Actions/Webhooks, GitLab (repo/pipeline storage), CI/CD Pipeline Design, Automated Testing, AI-Assisted Pipeline Scripting (via Claude)

Observability: CloudWatch (Alarms, Dashboards, Events), Prometheus, Grafana, Grafana Alerting, cAdvisor, Node Exporter, Blackbox Exporter, Spring Boot Actuator, AWS WAF Rule Sets

Databases: PostgreSQL (on-prem Docker + AWS RDS) – schema design, REST API-driven data modeling, MySQL, MongoDB, Redis, Schema Management, Query Optimisation, Docker-Volume Backup, S3 Disaster Recovery

Tooling: Git/GitHub, GitLab, Bitbucket, Jira, WinSCP, Apache, Linux CLI

PROFESSIONAL EXPERIENCE

Greatpelican Digital Systems Private Limited

Software Engineer (Cloud & DevOps Focus)

Apr 2024 – Jul 2026

Hyderabad, India

- Promoted from Intern to Full-Time Engineer within 4 months based on performance — one of the fastest promotions in the company. Sole engineer owning DevOps, cloud, and infrastructure end-to-end; collaborated with one other engineer on backend REST API development.
- **Cost Saved \$1,000/mo** — Architected a hybrid on-prem/AWS model (HP Z1 Tower + AutoSSH tunnel), cutting monthly AWS spend from \$1,200 to ~\$200 – an 83% reduction. Eliminated 6 standalone EC2 instances, consolidated all traffic through a single gateway. Later relocated the reverse-proxy gateway from AWS EC2 to Vultr (still via AutoSSH), then replaced AutoSSH entirely with Cloudflare Tunnel (cloudflared) for direct on-prem hosting; running uninterrupted in production for 12+ months.
- **Kafka Migration** — Replaced Confluent Kafka (managed) with self-hosted Apache Kafka, saving an additional \$300/mo while maintaining identical throughput and reliability for real-time EV session event streaming. Total infra savings: \$1,300+/mo.

- **Database Migration & Ownership** — Migrated production PostgreSQL databases off Amazon RDS onto a self-hosted Dockerised instance on-prem. Own end-to-end database operations: schema management, query optimisation, performance tuning, and automated Docker-volume backups to S3 for point-in-time disaster recovery. Also worked with MongoDB and deployed Redis as a containerised microservice for caching.
- **CI/CD & Automation** — Cut deployment cycle time by ~60% by building a master-slave Jenkins CI/CD architecture with GitHub Webhooks, Maven integration, and automated build-test-deploy pipelines across Dev, QA, and Production. Containerised 100% of application services with Docker, orchestrated on AWS ECS (Fargate & EC2). Used Terraform and Ansible, with AI-assisted scripting (via Claude) to codify environments quickly in a fast-paced, single-owner setup.
- **Security & Compliance** — Protected production EV charging APIs from DDoS and injection attacks via AWS WAF (zero security incidents since deployment); enforced IAM least-privilege, MFA, and SSH key-only auth; managed secrets via HashiCorp Vault; configured AWS Cognito (OAuth 2.0/JWT) with MFA across 3 production apps; implemented RSA asymmetric key-pair signing for sensitive API payloads.
- **Backend & Data** — Designed and maintained RESTful Spring Boot APIs for EV charging session management, travel expense tracking, and payment processing at production load; optimised self-hosted PostgreSQL for thousands of concurrent high-frequency writes with zero data loss; reduced API payload size and cut response times from ~700ms to under 200ms via caching, DTO projection, and response compression.
- **Observability** — Instrumented services and infrastructure with Prometheus, Grafana, cAdvisor, and Node Exporter for container- and host-level metrics, and Blackbox Exporter for endpoint/subdomain health checks, alongside CloudWatch dashboards and alarms; maintained 99.9%+ uptime via self-healing scripts and Spring Boot Actuator health checks integrated with ALB traffic shifting.

Greatpelican Digital Systems Private Limited

Jan 2024 – Mar 2024

Software Engineer Intern – Manual Testing & Backend

Hyderabad, India

- Joined as a manual tester and self-driven backend learner; quickly expanded beyond internship scope by proactively contributing to Java Spring Boot backend development – converted to full-time within 4 months.
- **Manual Testing** — Designed and executed test cases for EV charging APIs, payment flows, and user auth, building a strong understanding of production system behaviour from day one.
- **Backend Contribution** — Proactively learned and contributed to Java Spring Boot REST API development, going beyond the testing role to understand and fix backend issues.
- **Fast Learner** — Self-taught AWS fundamentals, Docker basics, and CI/CD concepts during the internship, directly enabling the transition to full infrastructure ownership post-promotion.

KEY PROJECTS

Chargebird | EV Charging & Payment Platform

Jan 2024 – Jul 2026

- Integrated Razorpay for end-to-end payment processing: one-time session charges, recurring subscription plans, automated billing cycles, and webhook-driven payment status handling — processing real-money transactions daily in production.
- Built the core financial logic layer: per-kWh and per-session charging cost computation, operator-to-owner revenue sharing splits, GST/tax calculations, and automated monthly statement generation.
- Implemented AWS RSA asymmetric key pairs for sensitive payload signing between mobile clients and backend APIs; S3 with lifecycle policies (auto-archive to Glacier) for station images, profile pictures, and receipts; Cognito (OAuth 2.0/JWT) auth with MFA and SNS notifications.
- Built master-slave Jenkins pipeline reducing deployment cycle time by ~60%, with GitHub Webhook triggers for fully automated test-and-deploy on every commit.

CSMS | Charging Station Management System

Jan 2024 – Jul 2026

- Implemented OCPP 1.6J over WebSockets for real-time bidirectional communication between CSMS backend and physical charging stations, enabling Remote Start/Stop, Smart Charging Profiles, and live kWh/voltage/session metrics.
- Integrated self-hosted Apache Kafka (migrated from Confluent) for high-throughput EV session event streaming, decoupling charging station events from downstream processing and saving \$300/mo.
- Optimised self-hosted on-prem Dockerised PostgreSQL for thousands of concurrent high-frequency session writes with zero data loss; reduced 3rd-party hardware integration time from weeks to under 1 day via comprehensive OCPP 1.6J API documentation.

Sandesh | Internal Email Microservice

Jan 2024 – Jul 2026

- Designed and built a standalone Spring Boot microservice solely responsible for all transactional email across all 4 production platforms (Chargebird, CSMS, Voyli, hybrid infra layer), decoupling email logic from every other service.

- Implemented JavaMail (SMTP) as primary delivery channel and AWS SES as failover/high-volume channel; template-driven REST API design covers OTP/MFA, payment confirmations, invoices, subscription notices, and fault alerts.

Voyli | Travel & Expense Management Platform

Jan 2024 – Jul 2026

- Implemented S3 Transfer Acceleration for global high-res travel image uploads; optimised PostgreSQL schema for complex split-payment and hotel metadata queries across multiple currencies.
- Containerised all services with Docker, enabling independent per-microservice scaling and zero-downtime deployments during feature releases.

Hybrid Cloud Integration | On-Prem to AWS + Vultr

Jan 2024 – Jul 2026

- Sole architect of the hybrid model: HP Z1 Tower (Ubuntu Server + Docker), originally connected via a persistent AutoSSH tunnel to an EC2 gateway; later relocated the gateway from AWS EC2 to Vultr (still via AutoSSH), before finally replacing AutoSSH with Cloudflare Tunnel (cloudflared) for zero-trust connectivity – eliminating the EC2 gateway entirely and reducing AWS usage to just Cognito and SES.
- Migrated production PostgreSQL from Amazon RDS to a Dockerised on-prem instance with automated Docker-volume backups to S3 for point-in-time disaster recovery; combined on-prem design (\$1,000/mo saved) with the Kafka migration (\$300/mo saved) for \$1,300+/mo total savings. Custom shell scripts monitor tunnel health and auto-restart on drops.

OCPI | EV Charging Roaming Interoperability (In Progress)

2026 – Jul 2026

- Building a standalone OCPI (Open Charge Point Interface) integration to enable roaming between the Charge Point Operator (CPO) network and external e-Mobility Service Providers (EMSPs), allowing users of other charging networks to access and pay for sessions on our stations. Independent project, currently in active development.

Personal Projects | Node.js

Personal

- Built and debugged personal Node.js projects, including a blogging platform and a weather app, outside of professional work.

EDUCATION

SRKR Engineering College (JNTUK)

2019 – 2023

B.Tech – Computer Science Engineering

Bhimavaram, India